

SAYAN SINHA

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EDUCATION

Indian Institute of Information Technology, Kottayam

B.Tech in Computer Science and Engineering (Cyber Security) | **CGPA: 7.54**

Kerala, India

Aug. 2023 – Apr. 2027

EXPERIENCE

Indian Institute of Information Technology Kottayam

May. 2026 – Present

Machine Learning Intern – UAV Behavioral Intrusion Detection Dataset

- Built an automated UAV behavioral data collection framework using PX4 SITL, Gazebo, MAVLink, and Linux syscall tracing to generate labeled datasets covering **19 distinct attack scenarios** alongside benign flight operations for intrusion detection research.
- Designed reproducible data pipelines for automated trace acquisition, metadata generation, attack validation, and dataset standardization, creating consistent behavioral datasets for scalable anomaly detection and cyberattack classification experiments.

HypeOn

Feb. 2026 – May. 2026

AI Engineer Intern

- Built AI-powered automation pipelines using Python, FastAPI, and Large Language Models (LLMs) to generate structured wallpaper metadata, enrich product attributes, and automate content preparation for production applications.
- Implemented automated AI workflows for wallpaper generation, room visualization, and image preprocessing by integrating LLMs, computer vision techniques, and scalable data processing pipelines into production products.

TECHNICAL SKILLS

Programming: Python, Java, C, C++, JavaScript, TypeScript, SQL

Machine Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, Random Forest, Decision Trees, Gradient Boosting, Regression, Classification, LSTM

Data Science: Pandas, NumPy, Feature Engineering, EDA, Statistical Modeling, Time Series Analysis (ARIMA), K-Means, Hierarchical Clustering

Computer Vision & NLP: OpenCV, MediaPipe, Transformers, Hugging Face

Generative AI: LangChain, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, Semantic Search, Embedding Models, FAISS, ChromaDB, Gemini API, OpenAI API, Ollama API

ML Deployment & APIs: FastAPI, Flask, Django, REST APIs

Databases: MySQL, PostgreSQL, MongoDB, Supabase, Firebase

Data Engineering & Visualization: ETL Pipelines, Data Pipelines, Data Cleaning, Power BI, Tableau, Excel

Tools: Git, GitHub, Jupyter Notebook, Joblib

PROJECTS

Football Player Market Value Prediction | *Python, Scikit-learn, XGBoost, Seaborn*

- Analyzed and engineered features from **100+ player attributes**, applying preprocessing, outlier treatment, and categorical encoding to improve regression model performance.
- Trained and evaluated multiple regression models, achieving **94.41% prediction accuracy** with XGBoost while outperforming Random Forest by nearly **9%**.

[GitHub](#)

FIA Formula 1 Real-Time Predictive Telemetry Engine | *Python, FastAPI, MongoDB, Power BI*

- Architected a real-time telemetry pipeline processing live Formula 1 race data at **3.7–5 Hz per car**, generating predictive features for tire degradation, energy deployment, and race strategy.
- Built FastAPI services backed by MongoDB and interactive Power BI dashboards to visualize live telemetry, pace deltas, and strategic race insights with low-latency updates.

AI Recipe-to-Grocery Matching Platform | *FastAPI, Python, Gemini API, Semantic Normalization*

- Created an NLP pipeline that transformed unstructured recipes into structured grocery lists using FastAPI, Gemini API, semantic normalization, and Supabase.
- Designed a persistent semantic ingredient cache that reduced LLM API costs by **40%** while improving response latency through ingredient deduplication.

[GitHub](#) | [Website](#)

Real-Time ASL Gesture Recognition System | *Computer Vision, MediaPipe, Scikit-learn, Joblib*

- Implemented a real-time gesture recognition system using MediaPipe and Random Forest, classifying **28 ASL gestures** with **99% accuracy**.
- Integrated a low-latency inference pipeline with operating system shortcuts, enabling seamless real-time gesture-controlled human-computer interaction.

[GitHub](#)

ACHIEVEMENTS

Security Data Analysis: Ranked **Top 4%** globally in picoCTF 2025. Team score: 4610. [Team Ranking](#)

Quantitative Skills: Solved **550+** algorithmic problems on LeetCode. [Profile](#)